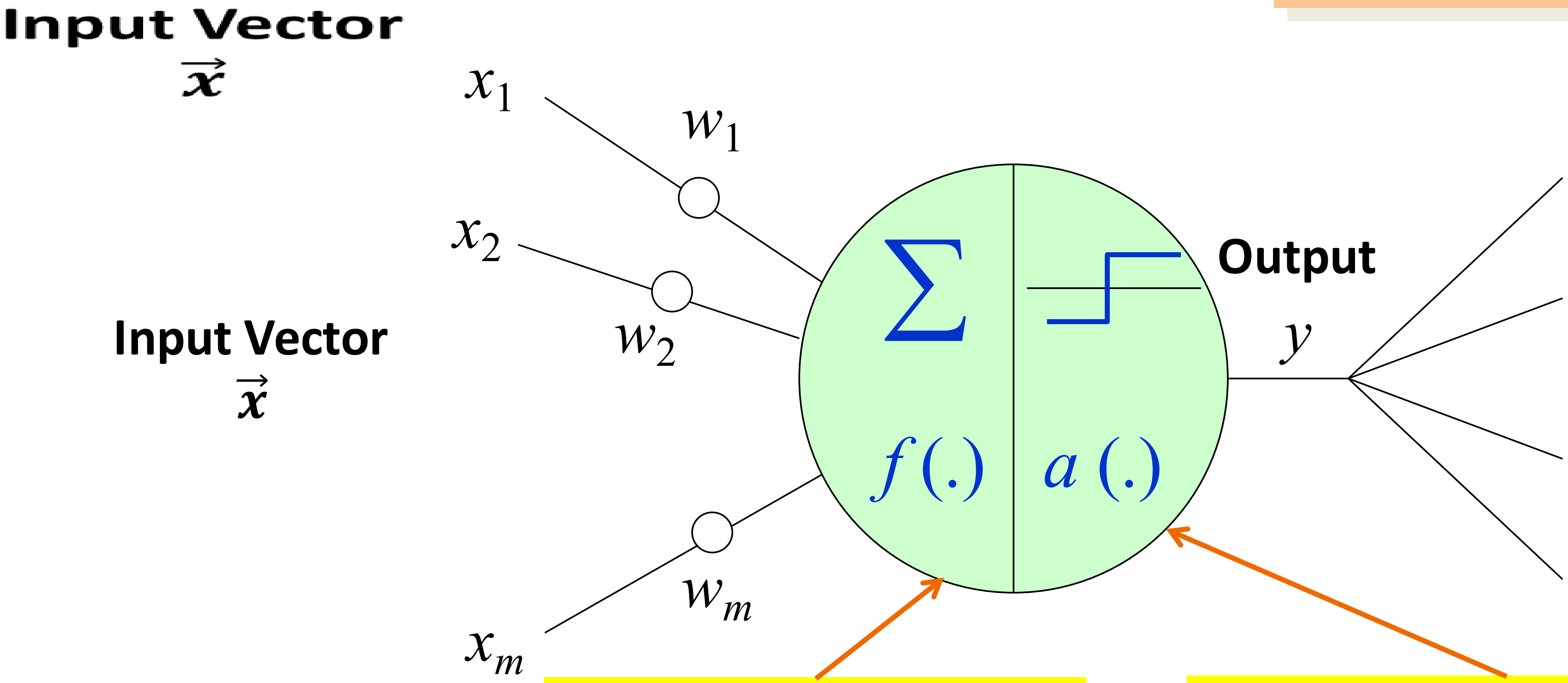
# Backpropagation

(Introduction)

# The Artificial Neuron

Proposed by McCulloch and Pitts [1943]
M-P neurons




**f**
**Integration Function**
***Aggregating input signals***

***a***
**Activation Function**
***Thresholding Logic***

# How to train a Perceptron?

Proposed by Rosenblatt [1962]

- Perceptron Algorithm - Iterative algorithm to learn the weight vector
- Basic Idea
  - Update weights in proportion to the error contributed by inputs

Randomly initialize weight vector $\vec{w}_0$

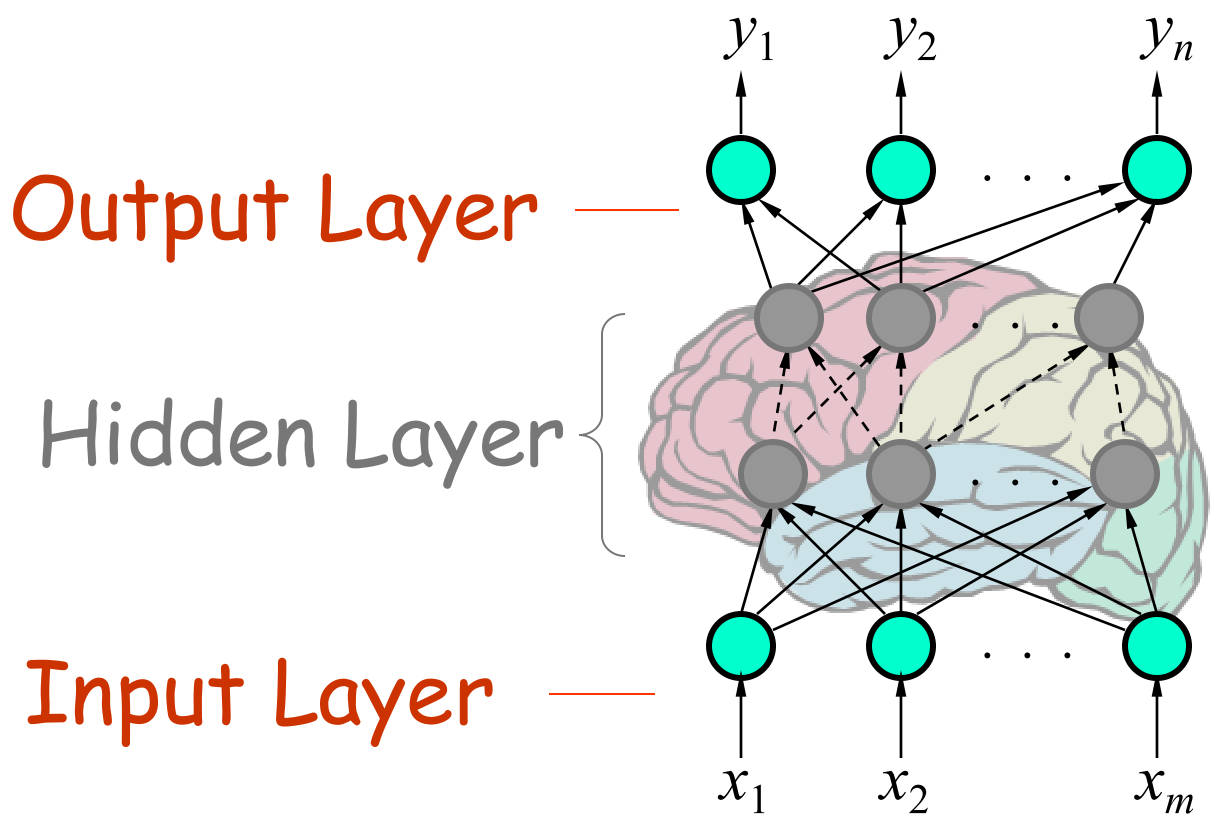
Repeat until *error is less than a threshold* $\gamma$ or max_iterations $M$:

For each training example $(\vec{x}_i, t_i)$:

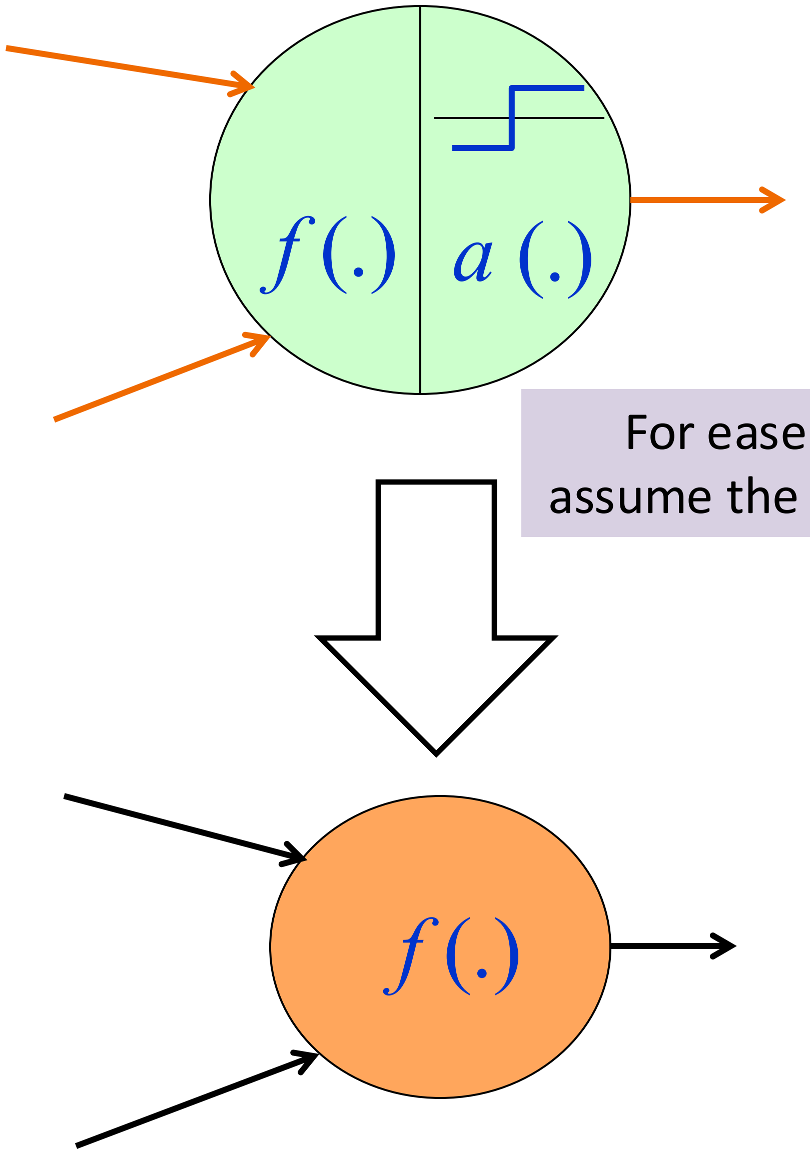
Predict output $y_i$ using current network weights $\vec{w}_n$
Update weight vector as follows:

$$\vec{w}_{n+1} = \vec{w}_n + \underbrace{\eta}_{\text{Learning Rate}} * \underbrace{(t_i - y_i)}_{\text{Error}} * \vec{x}_i$$

# Multi-Layered Perceptrons (MLPs)

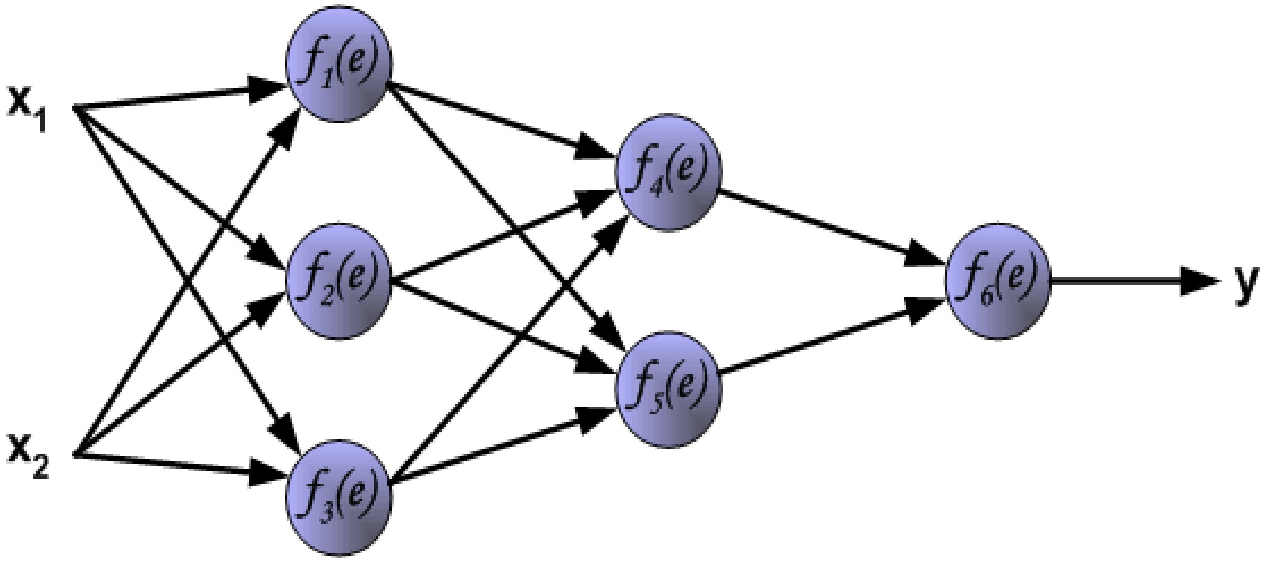


# The Back Propagation Algorithm




For ease of explanation, assume the change of notation

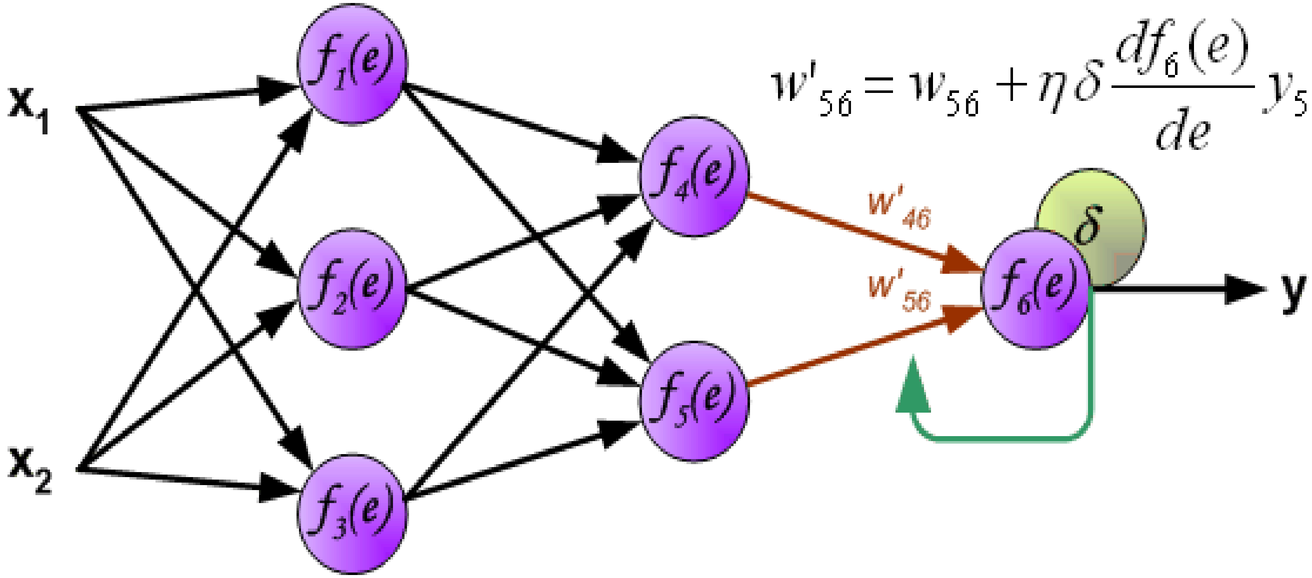
The resultant MLP



# Back Propagation Demo

$$w'_{46} = w_{46} + \eta \delta \frac{df_6(e)}{de} y_4$$

$$w'_{56} = w_{56} + \eta \delta \frac{df_6(e)}{de} y_5$$



# Convolution Layer

(Introduction)

# ImageNet

IMAGENET

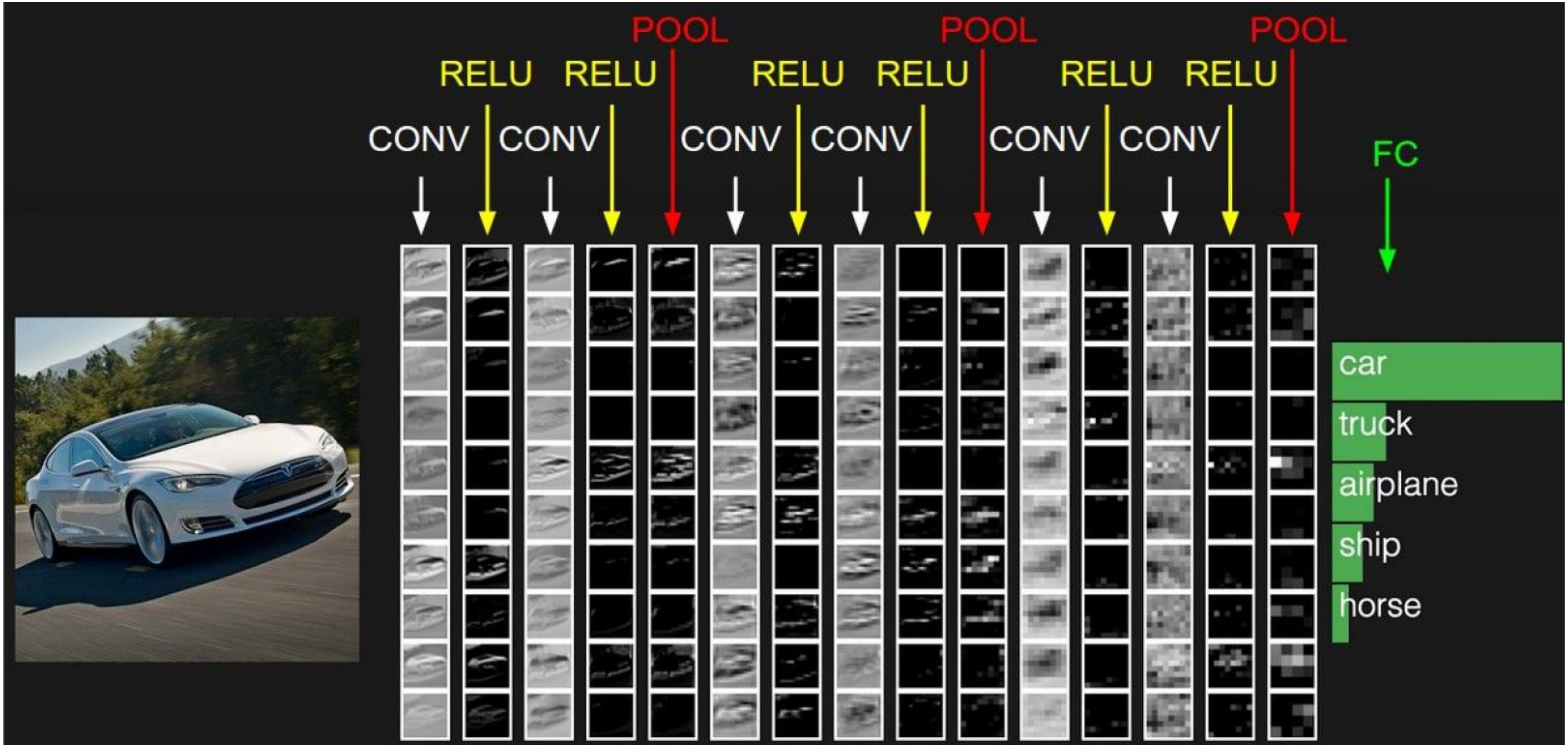
www.image-net.org

**22K** categories and **14M** images

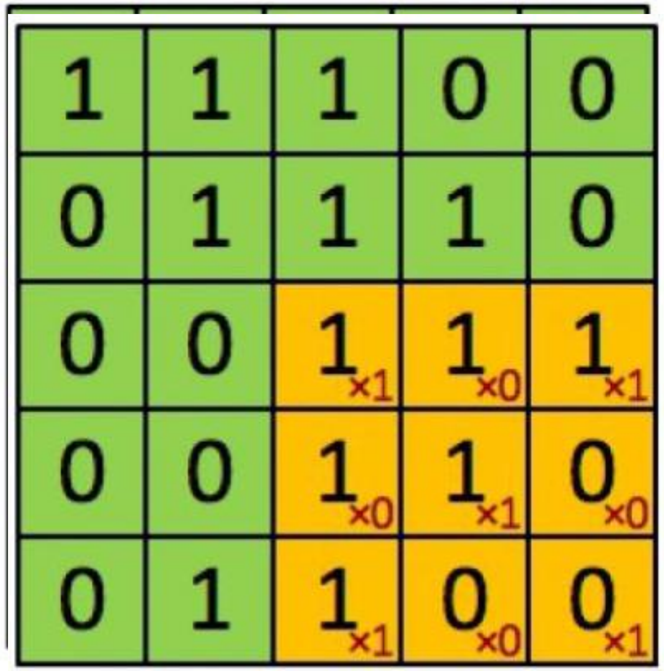
- Animals
  - Bird
  - Fish
  - Mammal
  - Invertebrate
- Plants
  - Tree
  - Flower
- Food
- Materials
- Structures
- Artifact
  - Tools
  - Appliances
  - Structures
- Person
- Scenes
  - Indoor
  - Geological Formations
- Sport Activities

http://image-net.org/download-faq

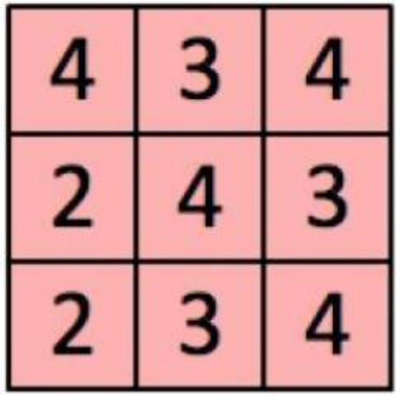
# ConvNet with Pooling and FC Layers



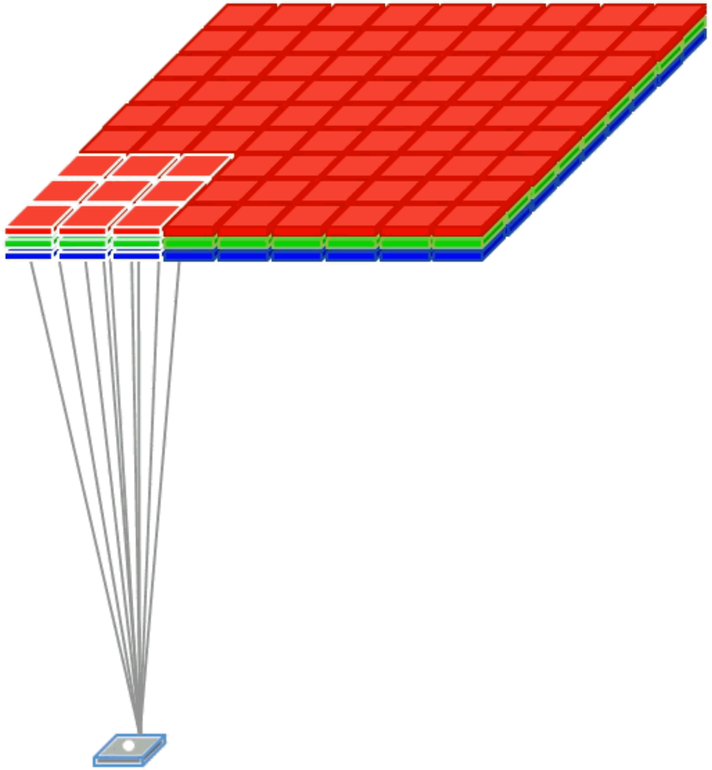
# What is convolution of an image with a filter?




Image




Convolved
Feature



# Thanks!!

**Questions?**